# Quadratic Functions & Complex Numbers Error Analysis

TEACHER KEY



Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

## 1. Vertex form: sign of *h*

**ORIGINAL PROBLEM**

Identify the vertex and the axis of symmetry of $f(x) = -(x + 2)^2 + 3$.

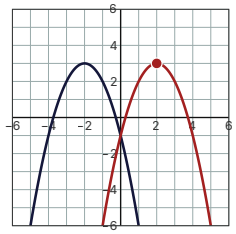
**INCORRECT WORKED SOLUTION**

Vertex form: $f(x) = a(x - h)^2 + k$

$f(x) = -(x + 2)^2 + 3$, so $h = 2$ and $k = 3$.

Vertex: (2, 3)

Axis of symmetry: $x = 2$



Navy: $f(x)$. Red: parabola with the student's vertex (2, 3).

**A. EXPLAIN THE MISTAKE (1 POINT)**

The vertex form has $(x - h)$. Here $x + 2 = x - (-2)$, so $h = -2$, not 2. The graph shows the student's vertex (2, 3) is not on the graph of $f$: $f(2) = -13$.

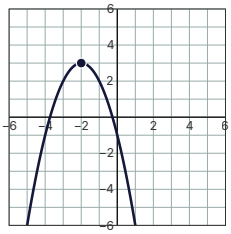
**B. SHOW CORRECT WORK (1 POINT)**

$x + 2 = x - (-2)$, so $h = -2$ and $k = 3$.

Vertex: $(-2, 3)$

Axis of symmetry: $x = -2$

Check: $f(-2) = -(0)^2 + 3 = 3$



Vertex $(-2, 3)$ on the graph of $f(x)$.

## 2. Completing the square

**ORIGINAL PROBLEM**

Write $f(x) = 2x^2 - 12x + 7$ in vertex form. Then state the minimum value.

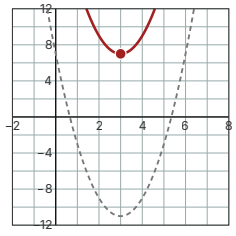
**INCORRECT WORKED SOLUTION**

$f(x) = 2(x^2 - 6x) + 7$

Half of $-6$ is $-3$, and $(-3)^2 = 9$:

$f(x) = 2(x^2 - 6x + 9) + 7$

$f(x) = 2(x - 3)^2 + 7$. Minimum value: 7



Dashed: $f(x)$. Red: the student's vertex form.

**A. EXPLAIN THE MISTAKE (1 POINT)**

Adding 9 inside the parentheses really adds $2 \cdot 9 = 18$ to the function, because of the factor 2 out front. The student never subtracted it. The red parabola (vertex (3, 7)) is not the graph of $f$: $f(3) = -11$.

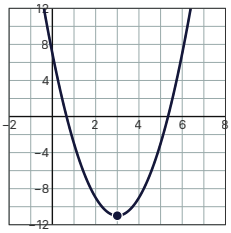
**B. SHOW CORRECT WORK (1 POINT)**

$f(x) = 2(x^2 - 6x + 9) + 7 - 2(9)$

$f(x) = 2(x - 3)^2 - 11$

Vertex $(3, -11)$; minimum value $-11$

Check: $f(3) = 18 - 36 + 7 = -11$



The vertex form matches $f(x)$. Vertex $(3, -11)$.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

# Quadratic Functions & Complex Numbers Error Analysis **TEACHER KEY**



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## 3. Square roots: the ± step

**ORIGINAL PROBLEM**

Solve $3(x + 1)^2 - 12 = 0$.

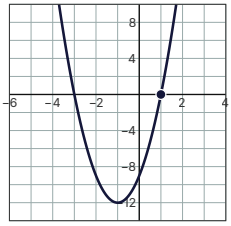
**INCORRECT WORKED SOLUTION**

$3(x + 1)^2 = 12$

$(x + 1)^2 = 4$

$x + 1 = 2$

$x = 1$




The graph of $y = 3(x + 1)^2 - 12$. Dot: the student's solution.

**A. EXPLAIN THE MISTAKE (1 POINT)**

Taking a square root gives TWO values, ±2, because both $2^2$ and $(-2)^2$ equal 4. The student kept only +2, so one $x$-intercept is missing from the graph.

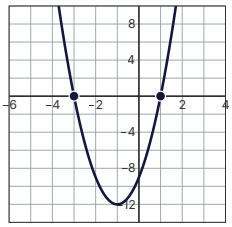
**B. SHOW CORRECT WORK (1 POINT)**

$(x + 1)^2 = 4$

$x + 1 = 2$ or $x + 1 = -2$

$x = 1$ or $x = -3$

Check: $3(-3 + 1)^2 - 12 = 0$




Two $x$-intercepts: $x = -3$ and $x = 1$.

## 4. Quadratic formula: dividing the numerator

**ORIGINAL PROBLEM**

Solve $3x^2 - 6x - 1 = 0$ with the quadratic formula.

**INCORRECT WORKED SOLUTION**

$a = 3, b = -6, c = -1$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(3)(-1)}}{2(3)} = \frac{6 \pm \sqrt{48}}{6}$$

$\sqrt{48} = 4\sqrt{3}$, so $x = \dfrac{6 \pm 4\sqrt{3}}{6}$

Divide by 6: $x = 1 \pm 4\sqrt{3}$

**A. EXPLAIN THE MISTAKE (1 POINT)**

The denominator 6 divides the ENTIRE numerator, not just the first term. Dividing both terms by 6 gives $1 \pm \dfrac{4\sqrt{3}}{6}$. Check the student's $x = 1 + 4\sqrt{3}$ in the equation: it gives 140, not 0.

**B. SHOW CORRECT WORK (1 POINT)**

$x = \dfrac{6 \pm 4\sqrt{3}}{6}$

Factor the numerator: $\dfrac{2(3 \pm 2\sqrt{3})}{6} = \dfrac{3 \pm 2\sqrt{3}}{3}$

$x = 1 \pm \dfrac{2\sqrt{3}}{3}$

Check: $3x^2 - 6x - 1 = 0$ for both values

# Quadratic Functions & Complex Numbers Error Analysis TEACHER KEY



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## 5. Discriminant with a negative *b*

**ORIGINAL PROBLEM**

Use the discriminant to decide how many real solutions $x^2 - 6x + 5 = 0$ has.

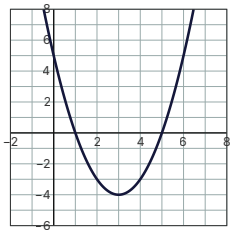
**INCORRECT WORKED SOLUTION**

$a = 1, b = -6, c = 5$

$b^2 - 4ac = -6^2 - 4(1)(5)$

$= -36 - 20 = -56$

The discriminant is negative, so there are no real solutions.



The graph of $y = x^2 - 6x + 5$.

**A. EXPLAIN THE MISTAKE (1 POINT)**

$b^2$ means $(-6)^2 = 36$, not $-6^2 = -36$. A negative number squared is positive, so you must put parentheses around $b$. The correct discriminant is positive, and the graph does cross the $x$-axis.

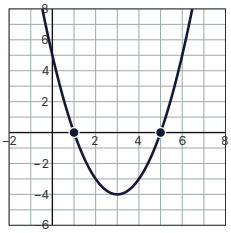
**B. SHOW CORRECT WORK (1 POINT)**

$b^2 - 4ac = (-6)^2 - 4(1)(5)$

$= 36 - 20 = 16$

$16 > 0$, so there are two real solutions.

Check: $x^2 - 6x + 5 = (x - 1)(x - 5)$, so $x = 1$ or $x = 5$



Two $x$-intercepts: 1 and 5.

## 6. Powers of *i*

**ORIGINAL PROBLEM**

Multiply $(2 + 3i)(1 - 4i)$ and write the product in the form $a + bi$.

**INCORRECT WORKED SOLUTION**

$2(1) + 2(-4i) + 3i(1) + 3i(-4i)$

$= 2 - 8i + 3i - 12i^2$

$= 2 - 5i - 12$

$= -10 - 5i$

**A. EXPLAIN THE MISTAKE (1 POINT)**

The student used $i^2 = 1$. But $i$ is defined so that $i^2 = -1$. So $-12i^2 = -12(-1) = +12$, not $-12$.

**B. SHOW CORRECT WORK (1 POINT)**

$2 - 8i + 3i - 12i^2$

$= 2 - 5i - 12(-1)$

$= 2 - 5i + 12$

$= 14 - 5i$

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## 7. Dividing complex numbers

**ORIGINAL PROBLEM**

Write $\frac{5 + 2i}{3 - i}$ in the form $a + bi$.

**INCORRECT WORKED SOLUTION**

Multiply the top and bottom by $3 - i$:

$\frac{(5 + 2i)(3 - i)}{(3 - i)(3 - i)} = \frac{15 - 5i + 6i - 2i^2}{9 - 3i - 3i + i^2}$

$= \frac{17 + i}{8 - 6i}$

Answer: $\frac{17 + i}{8 - 6i}$

**A. EXPLAIN THE MISTAKE (1 POINT)**

To clear $i$ from the denominator, multiply by the CONJUGATE $3 + i$ (change the sign of the imaginary part). Multiplying by the same $3 - i$ leaves an $i$ in the denominator, so the answer is not in $a + bi$ form.

**B. SHOW CORRECT WORK (1 POINT)**

Multiply by $\frac{3 + i}{3 + i}$: $\frac{15 + 5i + 6i + 2i^2}{9 - i^2}$

$= \frac{13 + 11i}{9 + 1} = \frac{13 + 11i}{10}$

$a + bi$ form: $\frac{13}{10} + \frac{11}{10}i$

Check: $(3 - i)(3 + i) = 10$, a real number

## 8. Quadratic inequality: which region?

**ORIGINAL PROBLEM**

Solve $x^2 - 2x - 8 > 0$. Write the solution in interval notation.

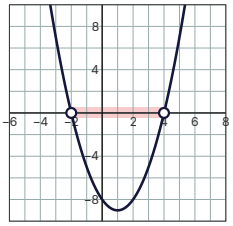
**INCORRECT WORKED SOLUTION**

Factor: $(x - 4)(x + 2) > 0$

Critical values: $x = 4$ and $x = -2$

Take the values between them: $-2 < x < 4$

Solution: $(-2, 4)$




Bar: the student's solution, on the graph of $y = x^2 - 2x - 8$.

**A. EXPLAIN THE MISTAKE (1 POINT)**

The parabola opens up. The inequality asks where $y > 0$, which is ABOVE the $x$-axis: outside the roots, not between them. Test $x = 0$ from the student's interval: $0 - 0 - 8 = -8$, and $-8 > 0$ is false.

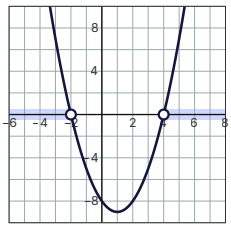
**B. SHOW CORRECT WORK (1 POINT)**

$y = x^2 - 2x - 8$ opens up, crossing the $x$-axis at $-2$ and 4.

$y > 0$ where the graph is above the $x$-axis: $x < -2$ or $x > 4$.

Open endpoints: the inequality is strict.

Solution: $(-\infty, -2) \cup (4, \infty)$




Bars: where the graph is above the $x$-axis.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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## 9. Quadratic inequality: shading

**ORIGINAL PROBLEM**

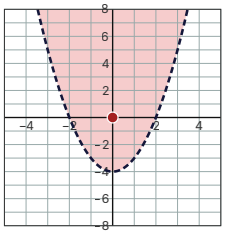
Graph $y < x^2 - 4$ and shade the solution set.

**INCORRECT WORKED SOLUTION**

Boundary: $y = x^2 - 4$, dashed because the inequality is strict.

The parabola opens up, so shade inside the U.

Shade the region above the curve.



Red region: student's. (0, 0) fails $y < x^2 - 4$.

**A. EXPLAIN THE MISTAKE (1 POINT)**

The graph is right, but the shading is not. $y <$ means $y$-values BELOW the curve. Test (0, 0), which is inside the U: $0 < 0 - 4$ is false. The solution is the region below the curve, outside the U.

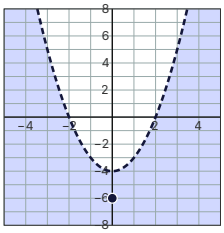
**B. SHOW CORRECT WORK (1 POINT)**

Boundary: dashed parabola $y = x^2 - 4$.

$y <$ means shade BELOW the curve (outside the U).

Test (0, −6): $-6 < 0 - 4$ is true.

Test (0, 0): $0 < -4$ is false, so it is not shaded.



Dashed boundary. Shade below the curve.

## 10. Maximum value of a quadratic

**ORIGINAL PROBLEM**

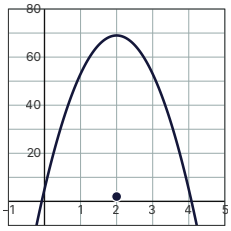
A ball is thrown upward. Its height is $h(t) = -16t^2 + 64t + 5$, where $t$ is in seconds and $h$ is in feet. Find the maximum height.

**INCORRECT WORKED SOLUTION**

The maximum occurs at the vertex.

$$t = -\frac{b}{2a} = -\frac{64}{2(-16)} = 2$$

The maximum height is 2 feet.



Dot: where the student's answer puts the ball at $t = 2$.

**A. EXPLAIN THE MISTAKE (1 POINT)**

The vertex has two coordinates. The formula gives the $t$-coordinate, the TIME of the maximum (2 seconds). The maximum HEIGHT is the output $h(2)$. The graph shows the ball is at 69 feet, not 2 feet, at $t = 2$.

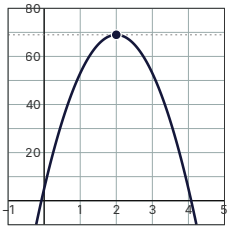
**B. SHOW CORRECT WORK (1 POINT)**

$t = 2$ seconds (the time of the maximum)

$h(2) = -16(2)^2 + 64(2) + 5$

$= -64 + 128 + 5 = 69$

Maximum height: 69 feet



Vertex (2, 69): $h = 69$ feet at $t = 2$ s.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points